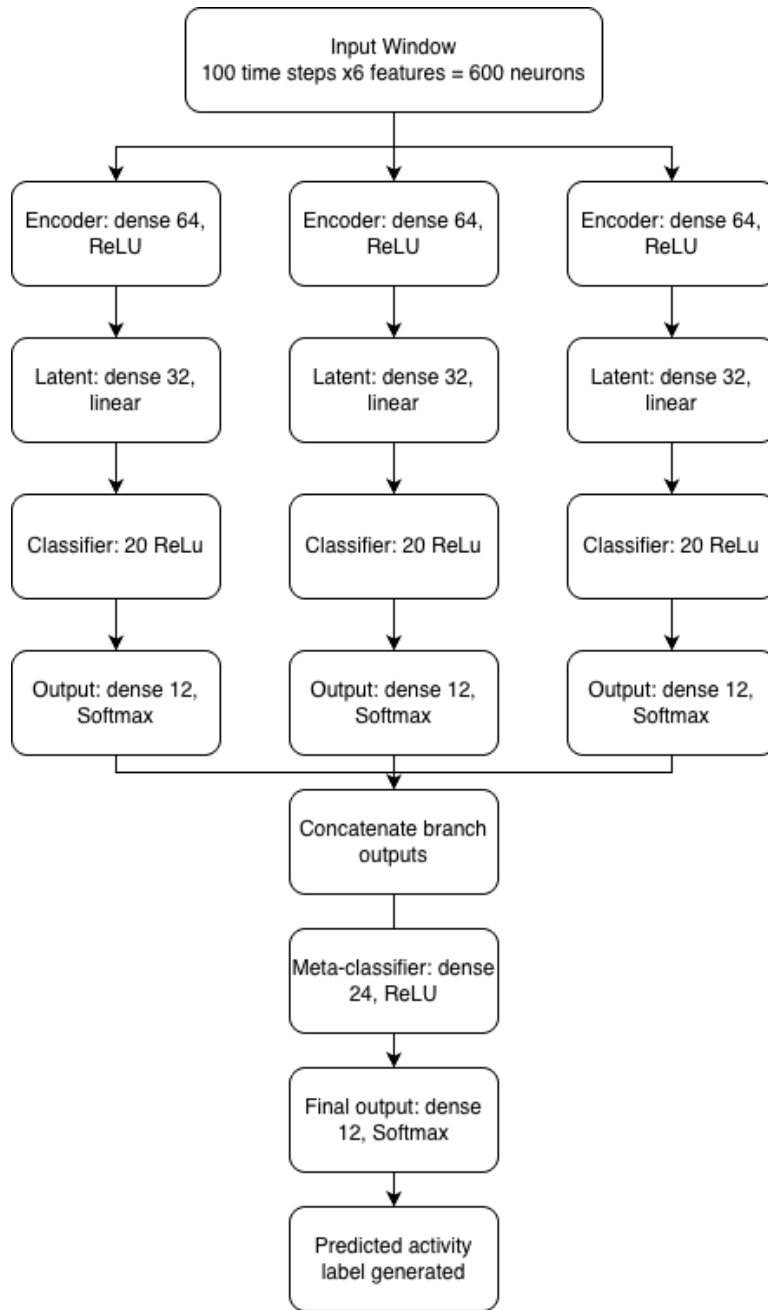


Part 1 Discussion:

The number of features used to train the Neural Network in Edge Impulse was 5. They were, 3 spectral features accX, accY, accZ, 2 output features, lift and idle. The number of input neurons is 600, as each window has 100 time steps and 6 IMU features. There are 3 branches: raw, standard-scaled, and min-max-scaled. Each branch has an encoder: Dense(64, ReLU) >> Dense(32, linear), and a classifier: Dense(20, ReLU) >> Dense(12, Softmax). The three 12-dimensional softmax outputs are concatenated into a 36-dimensional vector, which is passed into the meta-classifier: Dense(24, ReLU) >> Dense(12, Softmax)



Question 2:

1. **Pruning**, We used TF-MOT magnitude-based weight pruning with a PolynomialDecay schedule, increasing sparsity from 10% to 80% over 500 steps to remove low-importance weights and cut memory use.
2. **Quantization-Aware Training (QAT)**, We added fake quantization nodes during training to simulate INT8 inference while training in float, using `quantize_model()`.
3. **Stripping wrappers**, The pruning wrappers were stripped before QAT because `quantize_model()` cannot wrap already-wrapped layers.
4. **Mask enforcement**, A custom MaskEnforcerCallback was used to re-zero pruned weights after every batch, preventing QAT from recovering pruned weights.
5. **Hardware constraints**, The target device was the Arduino Nano 33 BLE Sense, which has around 1 MB flash / 256 KB SRAM, making size reduction essential.
6. **Final result**, INT8 quantization reduced model size significantly, improving inference speed and reducing power consumption on the Cortex-M4 CPU.

Question 3:

This is an acceptable output. When the board is resting flat, the model correctly predicts a static class with high confidence. This is a good result.

```
Sampling...
Predictions (DSP: 132.929993 ms., Class
#Classification results:
  Idle: 0.000000
  Lift: 0.996094
Starting inferencing in 2 seconds...
Sampling...
Predictions (DSP: 133.682999 ms., Class
#Classification results:
  Idle: 0.996094
  Lift: 0.000000
Starting inferencing in 2 seconds...
Sampling...
```

Question 4:

After deploying the ensemble model, the model repeatedly predicted the same class regardless of how the board was moved. This is likely because the mHealth dataset was collected with

sensors mounted on the body, so the gravity vector and axis orientation differ from the Arduino lying flat on a desk. Additionally, if the scaling parameters in the Arduino sketch do not exactly match those from the notebook, the inputs are out of distribution and the model cannot classify correctly.